

Numerical Methods for PDEs

by Andrea Cangiani

International School for Advanced Studies (SISSA)

September 2026

Contents

1	Well-posed problems and reliable discretizations	1
1.1	Why numerical methods?	1
1.2	Discrete problems	1
2	Elliptic tools: Green identities and maximum principles	4
2.1	Gauss–Green formulas	4
2.2	Harmonic functions	4
2.3	Poisson’s equation: uniqueness and stability	4
2.4	A general second-order operator	5
3	Centered finite differences in one dimension	7
3.1	The model problem and its discretization	7
3.2	The discrete maximum principle	7
3.3	A sharp stability estimate	8
3.4	Consistency and convergence	8
3.5	Conditioning and solver cost	9

1 Well-posed problems and reliable discretizations

1.1 Why numerical methods?

Analytical theory answers structural questions: Does a solution exist? Is it unique? How does it depend on the data? Even when those questions have satisfactory answers, a closed formula is rarely available. Moreover, the solution belongs to an infinite-dimensional function space. A numerical method replaces the continuous problem by a finite-dimensional problem whose solution can be computed.

Write an abstract problem as

$$\mathcal{P}(u, g) = 0, \quad (1.1)$$

where g collects the forcing, coefficients, domain, and boundary data. When the solution is unique, the solution operator is denoted by $u = R(g)$.

Definition 1.1 (Hadamard well-posedness). Problem (1.1) is well posed in specified data and solution norms if:

1. a solution exists for every admissible datum;
2. the solution is unique;
3. the solution depends continuously on the data.

In a locally Lipschitz setting, continuous dependence takes the form

$$\|R(g + \delta g) - R(g)\|_X \leq C_g \|\delta g\|_Y$$

for all sufficiently small admissible perturbations δg . For nonzero g and $R(g)$, a local relative condition number is

$$\kappa_R(g) = \limsup_{\varepsilon \downarrow 0} \sup_{0 < \|\delta g\|_Y \leq \varepsilon \|g\|_Y} \frac{\|R(g + \delta g) - R(g)\|_X / \|R(g)\|_X}{\|\delta g\|_Y / \|g\|_Y}.$$

A large condition number signals intrinsic sensitivity. *No numerical algorithm can remove sensitivity that already belongs to the continuous problem.*

1.2 Discrete problems

Let $h > 0$ denote a resolution parameter, usually a mesh width. For each h , let X_h be the finite-dimensional space of discrete unknowns and let Y_h be the space in which the discrete equations, residuals, and right-hand sides are measured. Both spaces are equipped with suitable norms. Once the discrete data g_h are fixed, the discrete operator acts as

$$\mathcal{P}_h(\cdot, g_h) : X_h \longrightarrow Y_h.$$

The numerical problem is

$$\mathcal{P}_h(u_h, g_h) = 0, \quad (1.2)$$

with $u_h \in X_h$. We first use the simplest notation, in which the exact solution can be inserted into the discrete equations and compared directly with u_h . The common situation in which sampling or interpolation is required is explained just after the basic error argument.

Definition 1.2 (Consistency). Let u be a sufficiently smooth exact solution, and let g_h denote the discrete representation of the exact data. The residual obtained by substituting u into the scheme is the *truncation error*

$$\tau_h = \mathcal{P}_h(u, g_h) \in Y_h.$$

The method is consistent if

$$\|\tau_h\|_{Y_h} \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Definition 1.3 (Stability). A family of discrete problems is stable if perturbations of the discrete data produce uniformly controlled perturbations of the discrete solution. For a linear operator $A_h : X_h \rightarrow Y_h$ and the problem $A_h u_h = f_h$, a standard form is

$$\|A_h^{-1}\|_{\mathcal{L}(Y_h, X_h)} \leq C \quad \text{with } C \text{ independent of } h.$$

Definition 1.4 (Convergence). A method is convergent if the numerical solution approaches the exact solution in the chosen solution norm:

$$\|u_h - u\|_{X_h} \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Proposition 1.5 (The basic error mechanism for linear problems). Suppose $A_h : X_h \rightarrow Y_h$, $A_h u_h = f_h$, and the exact solution can be inserted into the scheme so that $A_h u = f_h + \tau_h$. If $\|A_h^{-1}\| \leq C$ uniformly, then

$$\|u - u_h\|_{X_h} \leq C \|\tau_h\|_{Y_h}.$$

Thus consistency plus uniform stability implies convergence.

Proof. The error $e_h = u - u_h$ satisfies $A_h e_h = \tau_h$, so $e_h = A_h^{-1} \tau_h$. □

Sampling, interpolation, and the approximation property

The preceding notation deliberately suppresses a technical point. In many methods, u is a function whereas an element of X_h is a vector of nodal values, cell averages, or coefficients. The expression $A_h u$ and the difference $u_h - u$ are then not literally defined.

Introduce a sampling or interpolation operator

$$I_h : X \longrightarrow X_h.$$

The precise versions of consistency and discrete convergence are then

$$\|A_h I_h u - f_h\|_{Y_h} \longrightarrow 0, \quad \|u_h - I_h u\|_{X_h} \longrightarrow 0.$$

Stability gives the second statement from the first exactly as above. To conclude that the numerical method approximates the function u , one also requires an *approximation property*. If $X_h \subset X$, a typical assumption is

$$\|u - I_h u\|_X \longrightarrow 0.$$

With compatible norms, the triangle inequality then gives

$$\|u - u_h\|_X \leq \|u - I_h u\|_X + \|I_h u - u_h\|_X \longrightarrow 0.$$

If X_h consists only of grid vectors, use a reconstruction $R_h : X_h \rightarrow X$. The corresponding approximation property is $\|u - R_h I_h u\|_X \rightarrow 0$; if R_h is uniformly bounded, then

$$\|u - R_h u_h\|_X \leq \|u - R_h I_h u\|_X + C_R \|I_h u - u_h\|_{X_h} \longrightarrow 0.$$

Thus, in the fully precise formulation, approximation, consistency, and stability together imply convergence to the continuous solution.

Scope of the Lax equivalence theorem

The slogan *consistency plus stability implies convergence* is often associated with the Lax–Richtmyer theorem. That theorem is classically formulated for linear, well-posed initial-value problems and should not be quoted as an unrestricted theorem for every PDE discretization. This is in contrast with the Dahlquist Equivalence Theorem on numerical methods for ODEs.

Exercises

Exercise 1.1 (Absolute versus relative conditioning). Let $R(g) = g^{-1}$ for $g > 0$. Compute the absolute sensitivity $|R'(g)|$ and the relative condition number. Explain why the two quantities tell different stories near $g = 0$.

Exercise 1.2 (A scalar model). Consider $a_h u_h = f_h$ with $a_h > 0$, $a_h \rightarrow a > 0$, and $f_h \rightarrow f$. Identify a consistency error and a stability constant, and prove $u_h \rightarrow f/a$.

Exercise 1.3 (Residual versus error). Construct a 2×2 linear system for which a small residual does not imply a small error in the Euclidean norm. Relate the amplification to the condition number of the matrix.

Exercise 1.4 (Nonlinear extension). Suppose $F_h : X_h \rightarrow X_h$ is differentiable near a discrete solution and $F'_h(u_h)^{-1}$ is uniformly bounded. Use a first-order Taylor expansion to formulate a local analogue of the basic error mechanism.

2 Elliptic tools: Green identities and maximum principles

2.1 Gauss–Green formulas

Let $\Omega \subset \mathbb{R}^d$ be bounded with sufficiently regular boundary, and let n denote the outward unit normal. For $u \in C^1(\overline{\Omega})$, the divergence theorem gives

$$\int_{\Omega} \partial_i u \, dx = \int_{\partial\Omega} u n_i \, ds, \quad i = 1, \dots, d. \quad (2.1)$$

Consequently,

$$\int_{\Omega} \nabla u \, dx = \int_{\partial\Omega} u n \, ds, \quad (2.2)$$

$$\int_{\Omega} \Delta u \, dx = \int_{\partial\Omega} \partial_n u \, ds, \quad (2.3)$$

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = - \int_{\Omega} v \Delta u \, dx + \int_{\partial\Omega} v \partial_n u \, ds. \quad (2.4)$$

The last identity is Green’s first identity and is the bridge from strong to weak formulations.

2.2 Harmonic functions

A function $u \in C^2(\Omega)$ is harmonic if $\Delta u = 0$ in Ω . Harmonic functions satisfy the mean-value property:

$$u(x) = \int_{\partial B_r(x)} u \, ds = \int_{B_r(x)} \Delta u \, dy \quad \text{whenever } \overline{B_r(x)} \subset \Omega.$$

Conversely, a continuous function with the mean-value property is harmonic.

Theorem 2.1 (Maximum principles for harmonic functions). *Let Ω be connected and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic. Then*

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \quad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

If an extremum is attained at an interior point, then u is constant.

The strong statement follows from the mean-value property: if $u(x_0)$ equals the global maximum, then the average of u over every small ball about x_0 can equal that maximum only when u is constant on the ball. Connectedness propagates the conclusion.

Corollary 2.2 (Positivity). *If u is harmonic, $u = g \geq 0$ on $\partial\Omega$, and g is not identically zero, then $u > 0$ in Ω .*

2.3 Poisson’s equation: uniqueness and stability

Consider

$$-\Delta u = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega. \quad (2.5)$$

Corollary 2.3 (Uniqueness). *There is at most one classical solution of (2.5).*

Proof. The difference of two solutions is harmonic and vanishes on $\partial\Omega$; the maximum principle applied to w and $-w$ gives $w = 0$. $\square$

An equivalent energy proof uses (2.4):

$$0 = - \int_{\Omega} w \Delta w \, dx = \int_{\Omega} |\nabla w|^2 \, dx,$$

because $w = 0$ on the boundary.

Theorem 2.4 (A maximum-norm a priori estimate). *Suppose Ω is contained in a ball $B_R(x_0)$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ solves (2.5). Then*

$$\|u\|_{L^\infty(\Omega)} \leq \|g\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f\|_{L^\infty(\Omega)}.$$

Proof. Use the barrier $\phi(x) = (R^2 - |x - x_0|^2)/(2d)$, for which $-\Delta\phi = 1$ and $\phi \geq 0$ on Ω , and apply the maximum principle to suitable combinations of u , ϕ , and the boundary norm of g . $\square$

If $u, \tilde{u}$ correspond to data (f, g) and $(\tilde{f}, \tilde{g})$, the same estimate gives

$$\|u - \tilde{u}\|_{L^\infty(\Omega)} \leq \|g - \tilde{g}\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f - \tilde{f}\|_{L^\infty(\Omega)}.$$

This is continuous dependence, hence the stability part of well-posedness.

2.4 A general second-order operator

For clarity, write a linear operator in nondivergence form. Let $\Omega \subset \mathbb{R}^d$ be bounded and let

$$\mathcal{L}u = - \sum_{i,j=1}^d a_{ij}(x) \partial_{ij} u + \sum_{i=1}^d b_i(x) \partial_i u + c(x)u.$$

Assume that $A = (a_{ij})$ is symmetric and uniformly elliptic:

$$\lambda |\xi|^2 \leq \xi^\top A(x) \xi \leq \Lambda |\xi|^2, \quad \lambda > 0.$$

Theorem 2.5 (Weak maximum and comparison principles). *If $a_{ij}, b_i, c \in C(\overline{\Omega})$, with $c \geq 0$, then, for every $u \in C^2(\Omega) \cap C(\overline{\Omega})$:*

1. *if $\mathcal{L}u \leq 0$ in Ω , then*

$$\max_{\overline{\Omega}} u \leq \max \left\{ 0, \max_{\partial\Omega} u \right\};$$

if $c \equiv 0$, the zero may be omitted;

2. *if $\mathcal{L}u \leq \mathcal{L}v$ in Ω and $u \leq v$ on $\partial\Omega$, then $u \leq v$ in $\overline{\Omega}$.*

Remark 2.6. Analogous results hold for divergence-form operators $-\nabla \cdot (A \nabla u) + b \cdot \nabla u + cu$, but the natural weak hypotheses and proof are formulated in Sobolev spaces.

Exercises

Exercise 2.1 (Green's second identity). Derive

$$\int_{\Omega} (u \Delta v - v \Delta u) \, dx = \int_{\partial\Omega} (u \partial_n v - v \partial_n u) \, ds.$$

Exercise 2.2 (Maximum-principle uniqueness). Let $c \geq 0$. Prove uniqueness for $\mathcal{L}u = f$ with prescribed Dirichlet data by applying the comparison principle to the difference of two solutions.

Exercise 2.3 (Constructing a barrier). For $\Omega = (0, 1)^d$, construct a quadratic $\phi \geq 0$ satisfying $-\Delta\phi = 1$. Compare the resulting stability constant with the ball-based constant above.

Exercise 2.4 (Why $c \geq 0$?). Give a one-dimensional example showing that the maximum principle can fail when the zeroth-order coefficient is sufficiently negative.

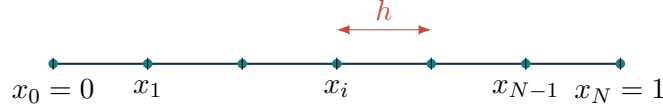
3 Centered finite differences in one dimension

3.1 The model problem and its discretization

Consider

$$-u'' = f \quad \text{in } (0,1), \quad u(0) = u(1) = 0. \quad (3.1)$$

Choose $N \in \mathbb{N}$, set $h = 1/N$, and define $x_i = ih$, $i = 0, \dots, N$.



The centered second difference is

$$\delta_h^2 v(x) = \frac{v(x+h) - 2v(x) + v(x-h)}{h^2}.$$

The finite-difference scheme seeks nodal values $U_i \approx u(x_i)$ such that

$$(\mathcal{L}_h U)_i := -\delta_h^2 U_i = f_i := f(x_i), \quad i = 1, \dots, N-1, \quad U_0 = U_N = 0. \quad (3.2)$$

In matrix form,

$$A_h \mathbf{U} = \mathbf{F}, \quad A_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

The matrix is symmetric positive definite.

3.2 The discrete maximum principle

Theorem 3.1 (Discrete maximum principle). *Let $V = (V_0, \dots, V_N)$. If $(\mathcal{L}_h V)_i \leq 0$ for $i = 1, \dots, N-1$, then*

$$\max_{0 \leq i \leq N} V_i = \max \{V_0, V_N\}.$$

If an interior maximum is attained and $\mathcal{L}_h V \leq 0$, then the maximum propagates through neighboring nodes; hence V is constant.

Proof. At an interior maximizer n ,

$$0 \leq -h^2(\mathcal{L}_h V)_n = V_{n+1} - 2V_n + V_{n-1},$$

whereas maximality gives $(V_{n+1} + V_{n-1})/2 \leq V_n$. Equality is forced, so both neighbors also attain the maximum. Repetition reaches the boundary. $\square$

3.3 A sharp stability estimate

Set $w(x) = x(1 - x)$ and $W_i = w(x_i)$. Since w is quadratic,

$$\mathcal{L}_h W_i = -w''(x_i) = 2, \quad W_0 = W_N = 0, \quad 0 \leq W_i \leq \frac{1}{4}.$$

For an arbitrary grid function V , apply the discrete maximum principle to

$$Z_i^\pm = \pm V_i - \frac{1}{2} \|\mathcal{L}_h V\|_{\ell^\infty} W_i.$$

This yields:

Lemma 3.2 (Maximum-norm stability). *For every grid function V ,*

$$\|V\|_{\ell^\infty} \leq \max\{|V_0|, |V_N|\} + \frac{1}{8} \|\mathcal{L}_h V\|_{\ell^\infty}. \quad (3.3)$$

In particular, the homogeneous discrete problem has only the zero solution. Since the system is square and finite dimensional, uniqueness also implies existence.

3.4 Consistency and convergence

For $u \in C^4([0, 1])$, Taylor's theorem with remainder gives

$$u''(x) - \delta_h^2 u(x) = -\frac{h^2}{24} \left(u^{(4)}(\xi_-) + u^{(4)}(\xi_+) \right), \quad (3.4)$$

where $\xi_- \in (x - h, x)$ and $\xi_+ \in (x, x + h)$. Therefore

$$\left| u''(x) - \delta_h^2 u(x) \right| \leq \frac{h^2}{12} \|u^{(4)}\|_{L^\infty(0,1)}.$$

Note that the remainder in (3.4) carries a minus sign when written as $u'' - \delta_h^2 u$. The sign does not affect the norm bound, but it matters in a pointwise asymptotic expansion.

Let $e_i = u(x_i) - U_i$. Since $e_0 = e_N = 0$,

$$(\mathcal{L}_h e)_i = (\mathcal{L}_h I_h u)_i - f_i = u''(x_i) - \delta_h^2 u(x_i).$$

Combining the truncation bound with (3.3) proves:

Theorem 3.3 (Second-order maximum-norm convergence). *If $u \in C^4([0, 1])$ solves (3.1) and U solves (3.2), then*

$$\max_{0 \leq i \leq N} |u(x_i) - U_i| \leq \frac{h^2}{96} \|u^{(4)}\|_{L^\infty(0,1)}. \quad (3.5)$$

The reusable proof pattern

$$\begin{array}{c} \boxed{\text{error equation}} + \boxed{\text{uniform stability}} \\ + \boxed{\text{truncation estimate}} \implies \boxed{\text{convergence estimate}}. \end{array}$$

3.5 Conditioning and solver cost

Spectral estimate for the Poisson matrix

The eigenvalues of A_h are

$$\lambda_k = \frac{4}{h^2} \sin^2 \left(\frac{k\pi}{2N} \right), \quad k = 1, \dots, N-1.$$

Hence

$$\text{cond}_2(A_h) \sim \frac{4}{\pi^2 h^2}.$$

Refinement improves approximation but worsens algebraic conditioning. This observation motivates preconditioners and multigrid methods.

Exercises

Exercise 3.1 (Exactness on quadratics). Prove that $\delta_h^2 p = p''$ for every polynomial of degree at most three. Find the error explicitly for $p(x) = x^4$.

Exercise 3.2 (A nonzero boundary condition). Derive the matrix system for $u(0) = \alpha$, $u(1) = \beta$. Show how the boundary values enter only the first and last entries of the right-hand side.

Exercise 3.3 (Energy stability). Multiply $(\mathcal{L}_h U)_i = f_i$ by hU_i , sum over the interior nodes, and prove a discrete integration-by-parts identity. Deduce a stability estimate in the discrete H^1 seminorm.

Exercise 3.4 (Observed order). Use the exact solution $u(x) = \sin(\pi x)$. Compute the discrete solution for successively halved h , and verify that $\|I_h u - U\|_{\ell^\infty}$ decreases by a factor close to four.